

Internet Café

Welcome to our group presentation

Terminal & Session Monitoring

- Real-Time Dashboard: A central view (PC View Controller) that displays the active/inactive status of all connected machines on the network.
- Remote Control: Administrators can remotely loc, unlock, restart, or shut down individual client terminals directly from the server interface.
- Session Tracking: Automated tracking of login times.

System Architecture & Tech Stack

- Frontend UI: Built with JavaFX for a responsive, modern administrative dashboard and client terminal interface.
- Backend Logic: Powered by Java, ensuring robust, multi-threaded performance for concurrent session handling.
- Database Management: MySQL integration via JDBC for secure, relational data storage (user profiles, transaction logs, admin credentials).
- Network Communication: RMI (Remote Method Invocation) connectivity for real-time synchronization between the main server and client PCs.

Problem Statement & Objective

- The Challenge: Managing multiple user sessions, tracking terminal status in real-time, and securely handling billing operations manually is inefficient and prone to errors.
- The Solution: A centralized multi-component software system designed to automate terminal monitoring, member authentication, and time-based billing. A centralized software system designed to automate terminal monitoring, handle member billing, and process food and beverage sales from a single interface.
- Core Objective: To provide a seamless, secure experience for users while giving administrators complete oversight and control over the network.

Member & Billing Logic

- Automated Conversion System: Custom billing logic that dynamically converts account balances in to screen time.
- Points-for-Time Exchange: A structured system where specific point values are deducted from a member's account to grant computer access (e.g., 5 points exchanged for 30 minutes of time).
- Guest vs: Member Access: Differentiated session rules for walk-in users versus registered members with stored balances.

Security & Authentication

- Secure Admin Gateway: Encrypt administer active login screens to prevent unauthorized access to system controls and financial data.
- OTP Verification Integration with Java Mail/SMTP libraries on handle secure One-Time Passwords (OTPs) sent directly to user emails for account recovery and identity verification.
- Data Integrity Transaction rollbacks and secure SQL querying to ensure member points and session data are never corrupted during unexpected system shutdowns.

Integrated Point of Sale (POS)

- Unified Inventory Management: A dedicated subsystem for tracking and selling physical goods like snacks, beverages, and cafe merchandise.
- Seamless Ordering: Customers can order items directly from their client terminals without leaving their seats, with orders instantly appearing on the admin dashboard.
- Flexible Payment Options: Integrated directly into the central MySQL database, allowing users to pay with cash at the counter or seamlessly deduct the cost from their existing member account balance/points.
- Transaction Synchronization: All F&B (Food and Beverage) sales are logged alongside PC usage data, providing administrators with a unified view of all revenue streams.

Future Enhancements & Conclusion

- Scalability: The modular design allows for easy addition of new terminals to the database without rewriting core logic.
- Potential Upgrades: Integration of automated report generation for daily revenue and peak usage times.
- Conclusion: This software provides a highly secure, automated, and scalable infrastructure, shifting the focus from manual tracking to optimized management.

Project Member

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Thank You